Quantum Mechanics II — Exercise Sheet 10

Bound states and the continuum

Variant: Solutions

Exercise 1 (Counting square-well levels at a threshold). For $V = -V_0 \mathbf{1}_{[-w,w]}$, put $\gamma = w\sqrt{2mV_0}/\hbar$. Use the lecture's even and odd matching equations to count strictly negative levels for $\gamma = \pi/2$, $\gamma = \pi$ and $\gamma = 5\pi/4$. Explain how a zero-energy threshold affects the count.

Solution. With $z = kw$ and $y = \eta w$, matching gives $z^2 + y^2 = \gamma^2$, with $y = z\tan z$ for even states and $y = -z\cot z$ for odd states. The even branches start at $z = j\pi$ and the odd branches at $z = (j + 1/2)\pi$. A branch gives a negative level exactly when its starting point is strictly below γ. Thus the counts are

γ	$\pi/2$	π	$5\pi/4$
negative levels	1	2	3

At a branch starting point equal to γ, one has $y = 0$. The exterior solution then fails to decay and is not a square-integrable zero-energy eigenvector. That threshold is not an additional bound level.

Exercise 2 (The deep well and its reference energy). Keep w fixed and let $V_0 \to \infty$ in the preceding well. For a fixed bound-state index $n \geq 1$, measured from the bottom, find the limit of $E_n(V_0) + V_0$. Explain why the unshifted $E_n(V_0)$ has no finite Dirichlet-box limit. Use the matching equations to justify the limiting wall condition.

Solution. Inside the well the kinetic energy is $E + V_0 = \hbar^2 k^2/(2m)$. For a fixed branch the exterior decay parameter tends to infinity as the well deepens. Matching requires $\tan(kw)$ or $-\cot(kw)$ to tend to positive infinity. The alternating even and odd roots therefore tend to $k_n = n\pi/(2w)$. It follows that

$$E_n(V_0) + V_0 \longrightarrow \frac{\hbar^2\pi^2 n^2}{8mw^2}.$$

This is the Dirichlet energy for an interval of length $2w$. The unshifted energies tend to $-\infty$ because their common reference bottom is $-V_0$.

Exercise 3 (A smooth trial state). For the square well take $R > 0$ and $f_R(x) = A_R/[1 + (x/R)^2]$, with $A_R > 0$ chosen to normalize it. (i) Compute A_R and $\langle f_R, Hf_R\rangle$. (ii) Show that this energy is negative for sufficiently large R for every $V_0 > 0$. (iii) Verify that $f_R \in D(H_0)$, so the lecture's variational principle applies.

Solution. The squared unnormalized norm is $\pi R/2$, so $A_R = \sqrt{2/(\pi R)}$. The function is smooth and it and its first two derivatives are square-integrable. Its Fourier characterization therefore gives $f_R \in D(H_0)$. Integration by parts and the substitution $y = x/R$ give

$$\|f_R'\|^2 = \frac{1}{2R^2}, \qquad \int_{-w}^{w} |f_R|^2 \,\mathrm{d}x = \frac{2}{\pi}\left(\arctan\frac{w}{R} + \frac{w/R}{1 + (w/R)^2}\right).$$

Consequently the trial energy is

$$\mathcal{E}(R) = \frac{\hbar^2}{4mR^2} - \frac{2V_0}{\pi}\left(\arctan\frac{w}{R} + \frac{w/R}{1+(w/R)^2}\right).$$

The leading potential term is $-4V_0 w/(\pi R)$, whereas the kinetic term has order R^{-2}. Thus the energy is negative for sufficiently large R. The variational principle and $\sigma_{\mathrm{ess}}(H) = [0,\infty)$ then imply a negative bound level.

Exercise 4 (An added repulsive potential). Let $V \geq 0$ be continuous and compactly supported, and $H = H_0 + V$ on $D(H_0)$. Show that H has no negative spectrum and that every $\lambda \geq 0$ lies in its spectrum. Use the lecture's shifted Weyl packets for the second claim.

Solution. For every $\psi \in D(H)$ the energy form is nonnegative:

$$\langle \psi, H\psi\rangle = \frac{\hbar^2}{2m}\|\psi'\|^2 + \int V|\psi|^2 \geq 0.$$

The lecture's bottom-of-spectrum result gives $\sigma(H) \subseteq [0,\infty)$. For $\lambda = \hbar^2 k^2/(2m)$ choose the normalized cut plane-wave packets of increasing width and shift each support entirely beyond the support of V. Their free residual tends to zero, while $V\psi_j = 0$ exactly. Thus $\|(H-\lambda)\psi_j\| \to 0$, including $k = 0$. Weyl's criterion gives $\lambda \in \sigma(H)$ and therefore $\sigma(H) = [0,\infty)$.